

lab04

1.

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

2.

```
fname = "WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

3.

```
countries = read_excel(fname, sheet = "Country")
```

4.

```
países_escolhidos = c("BRA", "USA", "RUS", "JPN", "DEU")
countries = countries %>%
  filter(`Country Code` %in% países_escolhidos)
```

5.

```
WDI = read_excel(fname, sheet = "Data")
```

6.

```
WDI = WDI %>%
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)
```

7.

```
WDI = WDI %>%
  pivot_longer(cols = `1960`:`2018`, names_to = "YEAR", values_to = "VALUE")
```

8.

```
class(WDI$YEAR)
```

```
[1] "character"
```

```
WDI = WDI %>%
  mutate(YEAR = as.integer(YEAR))
class(WDI$YEAR)
```

```
[1] "integer"
```

9.

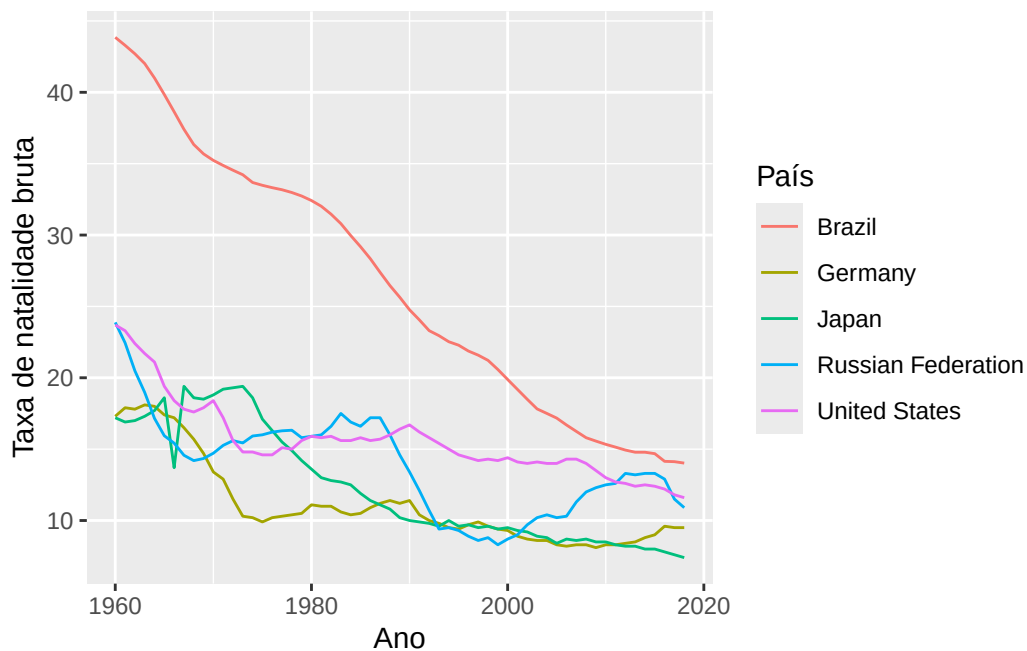
```
WDI = WDI %>%  
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") %>%  
  inner_join(countries %>% select(`Country Code`, `Short Name`), by = "Country Code") %>%  
  select(`Country Code`, `Short Name`, YEAR, VALUE) %>%  
  rename(SP.DYN.CBRT.IN = VALUE)
```

10.

```
WDI = WDI %>%  
  arrange(`Country Code`, YEAR)
```

11.

```
ggplot(WDI, aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +  
  geom_line() +  
  labs(x = "Ano", y = "Taxa de natalidade bruta", color = "País")
```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-09 19:40:25 -03"
```